Dimostrazione

- Progressione geometrica:

$$\sum_{k=0}^{n} x^k = \frac{1-x^{n+1}}{1-x}$$

$$P(0) = \sum_{k=0}^{0} x^k = x^0 = 1$$

$$\sum_{k=0}^{n+1} x^k = \frac{1-x^{(n+1)+1}}{1-x}$$

Assumo $P(n)$ e dimostro $P(n+1)$

$$\sum_{k=0}^{n+1} x^k = \sum_{k=0}^{n} x^k + x^{n+1} = \frac{1-x^{n+1}}{1-x} + x^{n+1} = \frac{1-x^{n+1}+x^{n+1}-x^{n+2}}{1-x} = \frac{1-x^{n+2}}{1-x} = \frac{1-x^{(n+1)+1}}{1-x}$$

- Binomio di Newton

$$P(n) = (a+b)^n = \sum_{k=0}^{n}\left(\binom{n}{k} a^{n-k} b^k\right)$$

$$P(0) = (a+b)^0 = \sum_{k=0}^{0}\left(\binom{0}{0} a^0 \cdot b^0\right) = 1$$

$$P(n+1) = (a+b)^{n+1} = (a+b)^n \cdot (a+b) = \sum_{k=0}^{n}\left[\binom{n}{k} a^{n-k} \cdot b^k\right] \cdot (a+b) =$$

$$= \sum_{k=0}^{n}\left[\binom{n}{k} a^{n+1-k} \cdot b^k + \binom{n}{k} a^{n-k} b^{k+1}\right] =$$

$$= \sum_{k=0}^{n}\left(\binom{n}{k} a^{n+1-k} \cdot b^k\right) + \sum_{k=0}^{n}\left(\binom{n}{k} a^{n-k} b^{k+1}\right) =$$

$$= \sum_{k=0}^{n}\left(\binom{n}{k} a^{n+1-k} \cdot b^k\right) + \sum_{k=1}^{n+1}\binom{n}{k-1} a^{n+1-k} \cdot b^k =$$

$$= a^{n+1} + \sum_{k=1}^{n}\left(\binom{n}{k} a^{n+1-k} \cdot b^k\right) + \sum_{k=1}^{n}\left[\binom{n}{k-1} a^{n+1-k} \cdot b^k\right] + b^{n+1}$$

$$= a^{n+1} + b^{n+1} + \sum_{k=1}^{n}\left[\left(\binom{n}{k} + \binom{n}{k-1}\right)\left(a^{n+1-k} \cdot b^k\right)\right]$$

$$= \sum_{k=1}^{n}\left[\binom{n+1}{k} a^{n+1-k} \cdot b^k\right] + a^{n+1} + b^{n+1}$$

$$= \sum_{k=0}^{n+1}\left[\binom{n+1}{k} a^{n+1-k} \cdot b^k\right] \qquad \text{Q.E.D}$$

1:

- Disuguaglianza di Bernoulli:

$$(1+x)^n \geq 1+nx, \quad x \geq -1 \wedge n \in \mathbb{N}$$

$P(0)$: $1 \geq 1$ vero

$P(n+1)$: $(1+x)^{n+1} = (1+x)^n \cdot (1+x) \geq (1+nx)(1+x) = 1+(n+1)x+nx^2 \geq 1+(n+1)x$ ($nx^2 \geq 0$)

$P(n+1) = (1+x)^n \geq 1+(n+1)x$

$$-\sum_{k=1}^{n} k^2 = \frac{1}{6}n(n+1)(2n+1), \quad x \geq -1 \wedge n \in \mathbb{N}$$

$$P(1) = \sum_{k=1}^{1} k^2 = \frac{1}{6}(1)(2)(3) = 1$$

$$P(n+1) = \sum_{k=1}^{n+1} k^2 = \frac{1}{6}(n+1)(n+2)(2n+2+1) = \frac{1}{6}(n+1)(n+2)(2n+3)$$

$$\sum_{k=1}^{n+1} k^2 = (n+1)^2 + \frac{1}{6}n(n+1)(2n+1) = \frac{6(n+1)^2 + n(n+1)(2n+1)}{6} = \frac{(n+1)(6(n+1)+n(2n+1))}{6} =$$

$$= \frac{n+1(6n+6+2n^2+n)}{6} = \frac{(n+1)(2n^2+7n+6)}{6} = \frac{(n+1)(n+2)(2n+3)}{6}$$

Q.E.D.

2:

2.1

$n! \geq 2^{n-1}, \forall n \in \mathbb{N}^+$

$P(1) = 1 \geq 1$

$P(n+1) = (n+1)! \geq 2^{n+1-1}$?

$1: (n+1)(n!) \geq 2^{n-1}(n+1) \xrightarrow{n+1 \geq 2, \forall n \in \mathbb{N}} (n+1)! \geq 2^n$

2.2

$$0 < a^n - b^n \leq (a+b)^{n-1}(a-b), \quad \forall a > b > 0 \text{ e } n \in \mathbb{N}^+$$

$P(1)$: $0 < a-b \leq (a+b)^0(a-b)$

$0 < a-b \leq a-b$

$P(n+1)$: $0 < a^{n+1} - b^{n+1} \leq (a+b)^n(a-b)$ — in prima.

? ? ?

3.

$$(1-x)^n \leq \frac{1}{1+nx}, \quad 0 < x < 1, \; n \in \mathbb{N}$$

$P(0) \Rightarrow 1 \leq 1 \rightarrow P(n+1) = (1-x)^{n+1} \leq \frac{1}{1+(n+1)x}$

$P(n+1) = (1-x)^{n+1} = (1-x)^n \cdot (1-x) \leq \frac{1}{1+nx} \cdot (1-x)$

$= (1-x)^n \cdot (1-x) \leq \frac{1-x}{1+nx} \Rightarrow \frac{1-x}{1+nx} \leq \frac{1}{1+(n+1)x}$

$= (1-x)(1+(n+1)x) \leq 1+nx =$

$= 1+(n+1)x - x - x^2(n+1) \leq 1+nx =$

$= \cancel{1}+nx \cancel{+x-x} - x^2(n+1) \leq \cancel{1}+nx =$

$= -x^2(n+1) \leq 0 =$

$= x^2(n+1) \geq 0 \rightarrow x > 0 \; \wedge n \geq 0$ Q.E.D.

4.

4.1

$$|a+b| \leq |a| + |b| =$$

$$(a+b)^2 \leq (|a| + |b|)^2 =$$

$$\cancel{a^2} + 2ab \cancel{+b^2} \leq \cancel{a^2+b^2} + 2|a||b|$$

$$2ab \leq 2|a||b|$$

$$ab \leq |a||b|$$

$$\{|ab|, ab\} \leq ab$$

$$\gamma = \frac{1+\sqrt{5}}{2}$$

$$\gamma + 1 = \gamma^2 \Rightarrow$$

$$\frac{1+\sqrt{5}}{2} + 2 = \left(\frac{1+\sqrt{5}}{2}\right)^2 \Rightarrow$$

$$\frac{3+\sqrt{5}}{2} = \left(\frac{6+2\sqrt{5}}{4}\right), \text{ Vero}$$

Dimostrazione per induzione doppia, che $F_n \geq \varphi^{n-2}$

$F_n = \varphi^{n-2}$, $\forall n \in \mathbb{N}$

$$P(0) = F_0 \geq \varphi^{-2} \rightarrow 1 \geq \varphi^{-2} \rightarrow 1 \geq \left(\frac{1+\sqrt{5}}{2}\right)^{-2} \rightarrow 1 \geq \left(\frac{2}{1+\sqrt{5}}\right)^2 \rightarrow 1 \geq \frac{2}{3+\sqrt{5}}$$

$P(n)$ vera

$$P(n+1) = F_{n+1} \geq \varphi^{n-1}$$

$$F_{n+1} = F_n + F_{n-1} \geq \varphi^{n-2} + F_{n-1}$$

$$\varphi^{n-2} + \varphi^{n-3} \rightarrow \varphi^{n-3}(\varphi+1) = \varphi^{n-3} \cdot \varphi^2 = \varphi^{n-1}$$

1.1

a1) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ → Vero

a2) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ → Falso

Assorbimento

b1) $A \cap (A \cup B) = A$ → Vero

b2) $A \cup (A \cap B) = A$ → Vero

1.2

Sia Ω un insieme, $\forall E \subseteq \Omega$,

$$X_E(\omega) = \begin{cases} 1, & \text{se } \omega \in E \\ 0, & \text{se } \omega \in \Omega \setminus E \end{cases}$$

Dimostrare che $\forall A, B \subseteq \Omega$

1) $X_{A \cap B}(\omega) = X_A(\omega)\, X_B(\omega)$?

1.3

i) $\neg(\forall q, P(q)) \rightarrow \exists q : \neg P(q)$

ii) $\neg(\exists q \text{ T.c. } P(q)) \rightarrow \nexists q \text{ T.c. } \neg P(q)$

iii) In quel quartiere, per ogni palazzo esiste un drago di parcheggio Tale che esiste una macchina parcheggiata in quell'area che non appartiene al residente del palazzo.

iv) esiste una ragazza in cui tutti in quel pub è non ballo quel ragazzo

1.4

1) $\sum_{k=0}^{3} (-1)^k (k+1) = 1 + (-2) + 3 + (-4) + 5 + (-6) + 7 + (-8) = -20 + 16 = -4$

2) $\sum_{k=1}^{4} k! = 1 + 2 + 6 + 24 = 33$

1.5

1.6

$$\sum_{k=1}^{n}\left(\frac{1}{k}-\frac{1}{k+1}\right)=\sum_{k=1}^{n}\left(\frac{1}{k(1+k)}\right)=\sum_{k=1}^{n}\left(k(1+k)\right)^{-1}$$

1.7

$$P_n=\sum_{k=1}^{2n}\frac{(-1)^{k+1}}{k}=\sum_{k=n+1}^{2n}\frac{1}{k}$$

$$P(1)=(-1)^2+\frac{(1)^3}{2}=\frac{1}{2}$$

$$1-\frac{1}{2}=\frac{1}{2}\Rightarrow \text{Vero}$$

$$P(n+1)=\sum_{k=1}^{2(n+1)}\frac{(-1)^{k+1}}{k}=\sum_{k=1}^{2n}\frac{(-1)^{k+1}}{k}+\frac{(-1)^{n+2}}{n+1}+\frac{(-1)^{n+3}}{n+2}$$

$$=\sum_{k=n+1}^{2n}\frac{1}{k}+\frac{(-1)^{n+2}}{n+1}+\frac{(-1)^{n+3}}{n+2}$$

? $\Rightarrow \sum_{k=n+2}^{2(n+1)}\frac{1}{k}$

1.9

$$(2-x)^4=16-32x+24x^2-8x^3+x^4$$

1.10

Progressione geometrica.

$$\sum_{k=0}^{n}x^k=\frac{1-x^{n+1}}{1-x},\quad \forall n\in\mathbb{N},\ \forall x\in\mathbb{R}\setminus\{1,0\}$$

$n\in\mathbb{N}$, $P(n)=\sum_{k=0}^{n}x^k=\frac{1-x^{n+1}}{1-x}$

- base induttiva:

$$\sum_{k=0}^{0}x^k=x^0=1 \quad // \quad \underset{\text{R.HS}}{\frac{1-x}{1-x}}=1$$

$\Rightarrow 1=1 \Rightarrow P(0)$ è vero

-) Supponi $P(n)$ è vero
-) $P(n+1)$? $\equiv \sum_{k=0}^{n+1}x^k=\frac{1-x^{n+2}}{1-x}$?

$$\sum_{k=0}^{n+1}x^k=x^{n+1}+\sum_{k=0}^{n}x^k \overset{P(n)}{=} x^{n+1}+\frac{1-x^{n+1}}{1-x}=\frac{x^{n+1}-x^{n+2}+1-x^{n+1}}{1-x}=\frac{1-x^{n+2}}{1-x}=\sum_{k=0}^{n+2}x^k$$

Q.E.D.

$\Rightarrow$ se $P(n)$ è vero $\Rightarrow P(n+1)$ è vero, per (P.I.) $P(n)$ è vero $\forall n\in\mathbb{N}$

Es 4

$|a| = \max\{a, -a\}$

① $|a+b| \le |a| + |b|$, ② $||a| - |b|| \le |a-b|$

- Per induzione $\left|\sum_{i=1}^{n} a_i\right| \le \sum_{i=1}^{n} |a_i|$

Dim:

- $|a| = \max\{a, -a\}$

$$\begin{array}{l} -|a| \le a \le |a| \\ -|b| \le b \le |b| \\ \hline -(|a|+|b|) \le a+b \le |a|+|b| \end{array} \quad (+) \quad \Rightarrow |a+b| \le |a| + |b|$$

$|a+b| = \max\{a+b, -a-b\}$

2) $|a| = |a-b+b| \overset{①}{\le} |a-b| + |b|$

$|b| = |b-a+a| \overset{②}{\le} |b-a| + |a|$ $\quad (|a-b|)$

$\Rightarrow |a| - |b| \le |a-b|$

$\Rightarrow |b| - |a| \le |a-b|$

$\Rightarrow \max\{|a|-|b|, -(|a|-|b|)\} \le (a-b)$

$\overset{\text{def}}{\Rightarrow} \overset{\text{di}}{\Rightarrow} \big||a|-|b|\big| \le |a-b|$

Esercizio 1:

1.1

$$\frac{2x^2-3}{x^4-5x^2+4} \xrightarrow{\alpha = x^2} \frac{2\alpha-3}{\alpha^2-5\alpha+4} = \frac{2\alpha-3}{(\alpha-1)(\alpha-4)} = \text{den} \neq 0 \rightarrow \alpha-1 \neq 0 \ \forall \ \alpha-4 \neq 0$$

$$x \neq \pm 1 \ \forall \ x \neq \pm 2$$

$$\text{Dom}(f) = \mathbb{R} \setminus \{-2, -1, 1, 2\}$$

1.2

$$x^4 - 4x^2 + 5 \xrightarrow{\alpha = x^2} \alpha^2 - 4\alpha + 5 = (\alpha-2)^2 + 1 \geq \rightarrow \text{Dom}(f) = \mathbb{R}$$

1.3

$$\frac{10^x}{2\sin(x)-1} \quad : \quad 2\sin(x) - 1 \neq 0 \rightarrow \sin(x) \neq \frac{1}{2} = x \neq \arcsin\left(\frac{1}{2}\right) \rightarrow x \neq \frac{\pi}{6} + k\pi$$

1.4

$$\frac{10^x - 100}{\sin(x) - 2} = \text{Dom}(f) = \mathbb{R}$$

1.5

$$\frac{1}{|x+2|\sqrt{x^2+2x-15}} \quad , \quad |x+2|\sqrt{x^2+2x-15} \neq 0 \qquad |x+2| \neq 0 \ \wedge \ \sqrt{x^2+2x-15} \neq 0$$

$$x \neq -2 \ \wedge \ (x+5)(x-3)$$

$$x \neq -2 \ \wedge \ x \neq -5 \wedge x \neq 3$$

1.6

$$\cos(\pi x^2) \qquad \text{Dom} \mathbb{R}$$

1.7

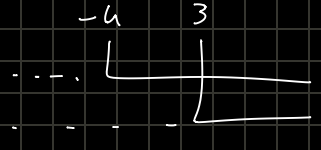
$$\text{Dom}(\pi x^2)$$

$$\pi x^2 \neq \frac{\pi}{2} \Rightarrow x^2 \neq \frac{1}{2} \Rightarrow x \neq \frac{\sqrt{2}}{2} + k\pi$$

1.8

$$\sqrt{-4^x + 2^x + 12} \xrightarrow{2^x = a} \sqrt{-x^2 + x + 12} \rightarrow x_{1,2} = \frac{1 \pm \sqrt{1 - 4(12)(-1)}}{-2} = \frac{1 \pm \sqrt{1+48}}{-2} = \frac{-1 \pm 7}{2} : \begin{matrix} -4 \\ 3 \end{matrix}$$

$$\sqrt{(x-3)(x+4)} \Rightarrow (x-3)(x+4) \geq 0 \Leftrightarrow x \in (-\infty, -4] \cup [3, +\infty)$$



1.9

$$|\arcsin(x)|^{1+\log_{\frac{1}{4}}|x|}$$

Dom: $\left(x \in [-1,1] \setminus \{0\}\right)$

1.10

$$\left(\arccos\left(2^x-3\right)\right)^{\sqrt{4^{2x}+2\cdot 4^x-3}} \overset{2^x=a}{\Longrightarrow} \left(\arccos\left(a-3\right)\right)^{\sqrt{a^4\cdot 2a^2-3}}$$

$$-1 \le 2^x-3 \le 1$$

$$2 \le 2^x \le 4$$

$$1 \le x \le 2 \qquad x \in [1,2]$$

Dom radice $\quad t=a^2$

$$a^4+2a^2-3 \ge 0 \Longleftrightarrow (T-1)(T+3) \ge 0 \Rightarrow T \le -3 \vee T \ge 1$$

$$4^x \le 3 \vee 4^x \ge 1$$

$$\nexists x \in \mathbb{R} \vee x \ge 0$$

$$\text{Dom}(f) = [1,2]$$

1.11

$$\arcsin\left(\frac{|x|}{1+|x|}\right) \rightarrow \text{Dom}(f) = \mathbb{R}$$

$$-\frac{\pi}{2} \le \arcsin\left(\frac{|x|}{1+|x|}\right) \le \frac{\pi}{2}$$

$$-1 \le \frac{|x|}{1+|x|} \le 1$$

$$\frac{|x|}{1+|x|} \le 1$$

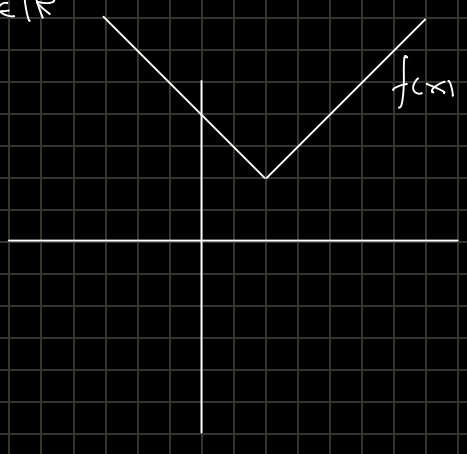
$$|x| \le 1+|x|$$

$$|x|-|x| \le 1+|x|-|x|$$

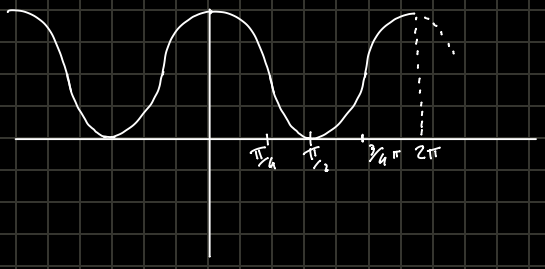
$$0 \le 1 \quad \forall x \in \mathbb{R}$$

ESERCIZIO 2.

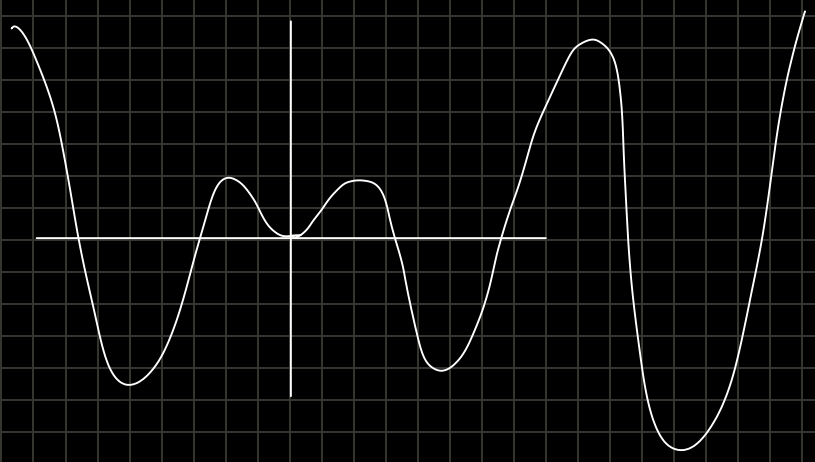
$$f(x) = |x-2|+2$$



$g(x) = 1 + \cos(2x)$



$h_1(x) = x \sin(x)$



Esercizio 3:

$h = f \circ g$, $K = g \circ f$

i) $f(x) = x + 1$, $g(x) = x^2 - 2x$

$h: f \circ g \to g(f(x)) = g(x+1) = (x+1)^2 - 2(x+1)$

$= x^2 + 2x + 1 - 2x + 2$

$= x^2 + 3$

ic) $f = x^2$ $\quad g(x) = \sqrt{x}$

$h: g \circ f = g(f(x)) = g(x^2) = \sqrt{x^2} = |x|$

$K: f \circ g = f(g(x)) = f(\sqrt{x}) = (\sqrt{x})^2 = |x|$

4

$f(x) = \sqrt[3]{x+1} - 2$

$Dom(f) = \mathbb{R}$

Iniettività:

$\sqrt[3]{x_1+1} - 2 = \sqrt[3]{x_2+1} - 2$

$\sqrt[3]{x_1+1} = \sqrt[3]{x_2+1}$

$x_1 + 1 = x_2 + 1 \Rightarrow x_1 = x_2$

$f'(x) = \frac{1}{3\sqrt[3]{(x+1)^2}}$ $\quad f'(x) \geq 0 \quad \forall x \in Dom(f') \to f$ monotona.

$x = \sqrt[3]{y+1} - 2 \rightarrow (x+2)^3 - 1 = y$

$f^{-1} = (x+2)^3 - 1$

Esercizio 5

Punti di accumulazione di $\mathbb{N}$ e $\mathbb{Z}$

in $\mathbb{N}$ $\nexists n \in \mathbb{N} : \forall I_{r(n)},\ I_{r(n)} \setminus \{n\} \cap \mathbb{N} \neq \emptyset$

in $\mathbb{Z}$ $\nexists z \in \mathbb{Z} : \forall I_{r(z)},\ I_{r(z)} \setminus \{z\} \cap \mathbb{Z} \neq \emptyset$

7 Usando la definizione di limite:

$$\lim_{x \to 0^+\infty} \frac{x}{1+x} = 1$$

Scheda 3:

esercizio 2:

$$\lim_{x \to 0} \frac{\sin(4x)}{3x}$$

$x \to 0^-$: $\frac{1}{0^-}$

$x \to 0^+$: $\frac{1}{0^+}$

$\Big\}$ $\lim_{x \to 0^-} f \neq \lim_{x \to 0^+} f \Rightarrow \nexists \lim_{x \to 0} f$

$$\lim_{x \to 0} \frac{e^{x^2} - 1}{x\sqrt{|x|}} = \frac{e^{x^2} - 1}{x^2 \cdot \sqrt{|x|}} \cdot \frac{\sqrt{|x|}}{\sqrt{|x|}} \Rightarrow 1 \cdot \sqrt{|x|} = 0$$

$x^2 \cdot x$

$$\left(\frac{f}{g}\right)'(x_0) \overset{?}{=} \frac{f'g - fg'}{g^2} \quad \text{Q.E.D.}$$

$$\left(f g^{-1}\right)'(x_0) \overset{②}{=} f' \cdot g^{-1} + f \cdot \left(g^{-1}\right)' \overset{③}{=} f' \cdot g^{-1} + f \cdot \frac{-g'}{g^2} = \frac{f'}{g} + \frac{-fg'}{g^2} = \frac{f'g - fg'}{g^2} \qquad g \neq 0$$

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases} \Longrightarrow \text{Verificare che non è derivabile in } x = 0$$

$$\{a_n\}_n := \frac{1}{2n\pi} \xrightarrow{n \to +\infty} 0$$

$$\{b_n\}_n := \frac{1}{\frac{\pi}{2} + 2n\pi} \xrightarrow{n \to +\infty} 0$$

$$\lim_{n \to +\infty} f(a_n) = 0 \neq 1 = \lim_{n \to +\infty} f(b_n)$$

$$\Longrightarrow f \notin C^0(0)$$

$D(\sin) = \cos(x)$?

$$\sin(x_1) - \sin(x_0) = 2\cos\left(\frac{x + x_0}{2}\right)\sin\left(\frac{x - x_0}{2}\right)$$

$$\lim_{x \to x_0} \frac{\sin(x) - \sin(x_0)}{x - x_0} = \lim_{x \to x_0} \frac{2\cos\left(\frac{x + x_0}{2}\right)\sin\left(\frac{x - x_0}{2}\right)}{x - x_0} \overset{\substack{h = \frac{x - x_0}{2} \Rightarrow x = 2h + x_0 \\ x \to x_0 \Rightarrow h \to 0}}{=}$$

$$= 2\cos(x_0 + h) \cdot \frac{\sin h}{2h} = \cos(x_0 + h)\left(\frac{\sin h}{h}\right) \xrightarrow{h \to 0} \cos(x_0) = 1$$

$$\sin(x) \sim x$$

Se $x_0 = 0$:

$$\lim_{x \to 0} \frac{\sin(x) - \sin(x_0)}{x - 0} \Rightarrow \lim_{x \to x_0} \frac{\sin(x)}{x} \overset{\downarrow}{=} 1$$

$D(\arccos(x))$ $\quad \arccos: [-1,1] \longrightarrow [\pi, 0]$

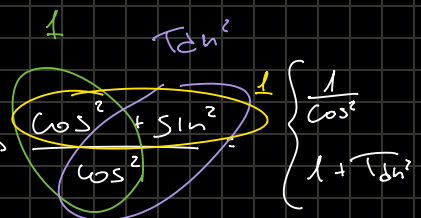
Sia $f(y) = \cos(y)\big|_{[\pi;0]} \Rightarrow f^{-1}(x) = \arccos(x)$ $\quad \sin^2+\cos^2=1 \Rightarrow \sin = \sqrt{1-\cos^2}$, $\cos = x_0$

$$D(f^{-1})(x_0) = \frac{1}{f'(f^{-1}(x_0))} = \frac{1}{\sin(\arccos(x_0))} = \frac{1}{\sin(y_0)} = \frac{1}{\sqrt{1-\cos^2(y_0)}} = \frac{-1}{\sqrt{1-x_0^2}}$$

($f'(f^{-1}(x_0))$: inversa; $\arccos(x_0) = y_0$)

$$D(\tan) = 1+\tan^2 = \frac{1}{\cos^2}$$

$$f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)} \Rightarrow f' = \frac{(\sin(x))'\cdot\cos(x) - \sin(x)\cdot(\cos(x))'}{\cos^2(x)} \Rightarrow$$



$$D(\arctan(1)) = \frac{1}{1+x^2}$$

Sia $f(y) = \tan(y)\big|_{(-\frac{\pi}{2},\frac{\pi}{2})} \Rightarrow f^{-1}(x) = \arctan(x)$ $\quad \arctan: (-\infty,+\infty) \longrightarrow [-\frac{\pi}{2},\frac{\pi}{2}]$

$$D(f^{-1}(x_0)) = \frac{1}{f'(f^{-1}(x_0))} = \frac{1}{1+\tan^2(\arctan(x_0))} = \frac{1}{1+x_0^2}$$

$$D(x^{x^x}) = y$$

$x^{x^x} = f(x) \Rightarrow x^x \log x = \log f(x)$

derivo il membro destro: $\frac{1}{f(x)}\cdot f'(x) = \frac{f'(x)}{f(x)}$

derivo il = sinistro: $(x^x\log x)' = (x^x)'\cdot\log x + (x^x\cdot\frac{1}{x}) \Rightarrow$

$\Rightarrow (e^{x\log(x)})'\cdot\log(x) + x^{x-1}$

$\Rightarrow e^{x\log(x)}\cdot(\log(x)+1)\cdot\log(x) + x^x\cdot\frac{1}{x}$

$\Rightarrow x^x\cdot(\log^2(x)+\log(x)+\frac{1}{x})$

$$\frac{f'(x)}{f(x)} = x^x\cdot(\log^2(x)+\log(x)+\frac{1}{x}) \Rightarrow f'(x) = x^{x^x}\cdot x^x\left[\log^2(x)+\log(x)+\frac{1}{x}\right]$$

$$= x^{x^x+x}\left[\log^2(x)+\log(x)+\frac{1}{x}\right]$$

Esercizio 3.2

Verificare:

$$\lim_{x\to -1} |9-2x| = 11$$

$\lim_{x\to x_0} f(x) = l \Rightarrow \forall \varepsilon>0\ \exists \delta>0$ t.c. $|x-x_0|<\delta \Rightarrow |f(x)-l|<\varepsilon$

$|x-(-1)|<\delta \Rightarrow |9-2x-11|<\varepsilon$?

fissato ε e risolvo per δ

$$-\varepsilon < -2x-2 < \varepsilon \Rightarrow \frac{\varepsilon}{2} > x+1 > -\frac{\varepsilon}{2} \Rightarrow |x+1| < \frac{\varepsilon}{2} \Rightarrow \delta = \frac{\varepsilon}{2}$$

prendo $\delta \le \frac{\varepsilon}{2}$

- Verificare $\lim_{x\to 2} |x^2+4| = 8$

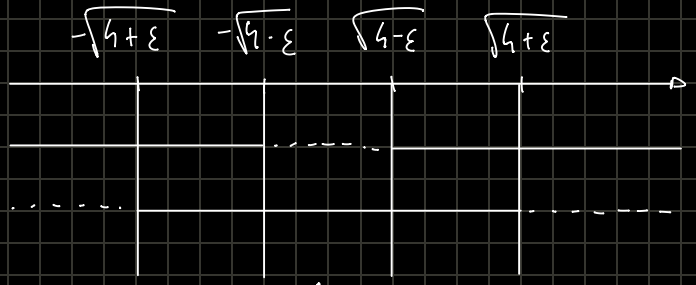
$|x-2|<\delta \Rightarrow |x^2+4-8|<\varepsilon$?

fissato ε e risolvo per δ:

$$-\varepsilon < x^2-4 < \varepsilon \Rightarrow \begin{cases} x^2-4 > -\varepsilon & ① \\ x^2-4 < \varepsilon & ② \end{cases}$$

① $x^2-4+\varepsilon>0 \Rightarrow x^2-4+\varepsilon=0 \Rightarrow x=\pm\sqrt{4-\varepsilon} \Rightarrow x<-\sqrt{4-\varepsilon} \wedge x>\sqrt{4-\varepsilon}$

② $x^2-4-\varepsilon<0 \Rightarrow x=\pm\sqrt{4+\varepsilon} \Rightarrow -\sqrt{4+\varepsilon}<x<\sqrt{4+\varepsilon}$



$$x \in \left(-\sqrt{4+\varepsilon}, -\sqrt{4-\varepsilon}\right) \cup \left(\sqrt{4-\varepsilon}, \sqrt{4+\varepsilon}\right)$$

Sapendo che $2 \in \left(\sqrt{4-\varepsilon}, \sqrt{4+\varepsilon}\right) \Rightarrow |x-2| \in \left(\sqrt{4-\varepsilon}-2, \sqrt{4+\varepsilon}-2\right)$

$$\delta = \min\left\{2-\sqrt{4-\varepsilon},\ 2-\sqrt{4+\varepsilon}\right\}$$

3.3

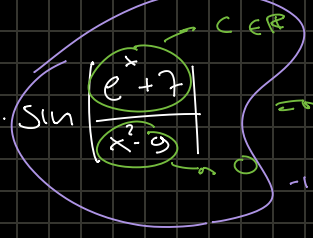
1) $\lim_{x\to 1} \frac{\sin(2\pi x)}{x-1} \overset{y=x-1}{\underset{x\to 1,\, y\to 0}{=}} \lim_{y\to 0} \frac{\sin(2\pi(y+1))}{y} = \lim_{y\to 0} \frac{\sin(2\pi y + 2\pi)}{y} = \lim_{y\to 0} \frac{\sin(2\pi y)}{2\pi y} \cdot 2\pi = 2\pi$ (lim notevole)

2) $\lim_{x\to 0} \frac{1-\cos(x)}{x} = \lim_{x\to 0} \frac{1-\cos(x)}{x^2} \cdot x = \frac{1}{2} \cdot 0 = 0$

3) $\lim_{x\to \frac{\pi}{2}} \frac{1-\sin x}{(x-\frac{\pi}{2})^2} \overset{y=x-\frac{\pi}{2}}{\underset{x\to\frac{\pi}{2},\, y\to 0}{=}} \lim_{y\to 0} \frac{1-\sin\left(y+\frac{\pi}{2}\right)}{y^2} = \frac{1-\cos y}{y^2} = \frac{1}{2}$ 4) = 1 lim not.

3.4

1) $\lim_{x\to 3} (x-3)\cdot \sin\left(\frac{e^x+7}{x^2-9}\right) = \lim_{x\to 3} (x-3)\cdot \sin\left(\frac{e^x+7}{x^2-9}\right) \Rightarrow$ limite · infinitesimo $= 0$



2) $\lim_{x\to 0} \sqrt[3]{x}\ \operatorname{arctan}\left(\frac{e^{-x}+7\log(x^2+9)}{x^5}\right) \Rightarrow$ infinitesimo per limitato $\Rightarrow 0$

$\lim_{x\to\infty} \frac{e^{-x}+7\log(x^2+9)}{x^5} = \frac{e^{-x}+7\log(9)}{x^5} \Rightarrow \frac{1+7\log 9}{x^5} \xrightarrow{x\to\infty} \frac{+\infty}{+\infty}$

3.5:

1) $\lim_{x\to+\infty}\left(e^x \cdot \frac{x}{x+1}\right) = \lim_{x\to+\infty} e^x + \lim_{x\to+\infty} 1 - \frac{1}{x+1} = +\infty + 1 - 0 \Rightarrow +\infty$

2) $\lim_{x\to+\infty}\left(-x^2 - e^{-\sqrt{x}}\right) = -\infty + 0 = -\infty$

3.6:

1) $\lim_{x\to+\infty}(x+\sin(x))$ Teo confr: $\Rightarrow -1 \le \sin(x) \le 1 \Rightarrow x-1 \le x+\sin(x) \le x+1 \xrightarrow{x\to+\infty} +\infty \le x+\sin(x) \le +\infty \Rightarrow \lim_{x\to+\infty} x+\sin(x) = +\infty$

2) $\lim_{x\to 0^+} \frac{3+\cos\left(\frac{1}{x}\right)}{x}$ Teo confr $\Rightarrow$ $-1 \le \cos\left(\frac{1}{x}\right) \le 1 \Rightarrow +2 \le \cos\left(\frac{1}{x}\right)+3 \le 4 \Rightarrow \frac{2}{x} \le \frac{3+\cos\left(\frac{1}{x}\right)}{x} \le \frac{4}{x}$

$\frac{2}{x} \to +\infty$; $\frac{3+\cos\left(\frac{1}{x}\right)}{x} \to \infty$; $\frac{4}{x} \to +\infty$

$\lim_{x\to 0^+} \frac{3+\cos\left(\frac{1}{x}\right)}{x} = +\infty$

3.7:

1) $\lim_{x\to-\infty}\left(\frac{x^3}{3x^2-4} - \frac{x^2}{3x+2}\right) = \lim_{x\to-\infty}\left(\frac{x^3(3x+2)-x^2(3x^2-4)}{(3x^2-4)(3x+2)}\right) = \lim_{x\to-\infty}\left(\frac{3x^4+2x^3-3x^4+4x^2}{9x^3+6x^2-12x-8}\right) =$

$\lim_{x\to-\infty} \frac{x^3\left(2+\frac{4}{x}\right)}{x^3\left(9+\frac{6}{x}-\frac{12}{x^2}-\frac{8}{x^3}\right)} = \frac{2}{9}$

2) $\lim_{x\to 2} \frac{2x^3-5x^2-4x+12}{x^4-4x^3+5x^2-4x+4} = \frac{0}{0}$ "Teo Hop. $\Rightarrow \frac{6x^2-10x-4}{4x^3-12x^2+10x-4}$ hopital $\Rightarrow \frac{12x-10}{12x^2-24x+10} = \frac{14}{10} = \frac{7}{5}$

3)

Scheda 5:

$$f(x)=\frac{x+\sqrt{x}}{x-\sqrt{x}}=\frac{x+\sqrt{x}-\sqrt{x}+\sqrt{x}}{x-\sqrt{x}}=1+\frac{2\sqrt{x}}{x-\sqrt{x}}$$

$$\left(\frac{f}{g}\right)'=\frac{(x+\sqrt{x})'(x-\sqrt{x})-(x+\sqrt{x})(x-\sqrt{x})'}{(x-\sqrt{x})^2}$$

$$=\frac{\left(1+\frac{1}{2\sqrt{x}}\right)(x-\sqrt{x})-(x+\sqrt{x})\left(1-\frac{1}{2\sqrt{x}}\right)}{(x-\sqrt{x})^2}=\frac{x-\sqrt{x}+\frac{x}{2\sqrt{x}}-\frac{\sqrt{x}}{2\sqrt{x}}-x+\sqrt{x}+\frac{\sqrt{x}}{2\sqrt{x}}}{(x-\sqrt{x})^2}$$

$$=\frac{-\frac{1}{2}\frac{x}{\sqrt{x}}-\frac{1}{2}\frac{x}{\sqrt{x}}}{(x-\sqrt{x})^2}=\frac{-\frac{x}{\sqrt{x}}}{(x-\sqrt{x})^2}=\frac{-\sqrt{x}}{(x-\sqrt{x})^2}$$

$$f(x)=\frac{1+\cos^2(x)}{2\sin(x)}=\frac{2-\sin^2(x)}{2\sin(x)}=\frac{2}{2\sin(x)}-\frac{\sin^2(x)}{2\sin(x)}=\frac{1}{\sin(x)}-\frac{\sin(x)}{2}$$

$$f'(x)=\frac{-\cos(x)}{\sin^2(x)}-\frac{1}{2}\cos(x)$$

$$h(x)=e^{\sqrt{x}}\Rightarrow h'(x)=e^{\sqrt{x}}\cdot\frac{1}{2\sqrt{x}}=\frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

$$z(x)=\sin\left(\sqrt{x^2+1}\right)$$

$$\int_0^2|x^2-1|\,dx\Rightarrow|x^2-1|\geq 0\Rightarrow x^2-1\geq 0\Rightarrow x\leq -1\ \vee\ x\geq 1$$

$$\Rightarrow|x^2-1|=\begin{cases}x^2-1, & x\geq 1\\ -x^2+1, & -1<x<1\\ x^2-1, & x\leq -1\end{cases}$$

$$\Rightarrow\int_0^2|x^2-1|\,dx=\int_0^1-x^2+1\,dx+\int_1^2x^2-1\,dx=$$

$$=\left[-\frac{x^3}{3}+x\right]_0^1+\left[\frac{x^3}{3}-x\right]_1^2$$

$$=-\frac{1}{3}+1-0+\frac{8}{3}-2-\frac{1}{3}+1$$

$$=\frac{6}{3}=2$$

$$\int_1^{+\infty}\frac{1}{1+x^2}\,dx=\lim_{b\to+\infty}\int_1^b\frac{1}{1+x^2}\,dx=\lim_{b\to+\infty}\left[\arctan(x)\right]_1^b=\lim_{b\to+\infty}\arctan(b)-\frac{\pi}{4}$$

$$=\frac{\pi}{2}-\frac{\pi}{4}=\frac{\pi}{2}$$

$$\lim_{x\to 0} \frac{x\int_0^x (\cos t)^3\,dt}{1-\cos(x)}$$

$$\int_0^x (1-\sin^2(T))\cos(T)\,dT = \int_0^{\sin(x)} 1-u^2\,du = u-\frac{u^3}{3}\Big|_0^{\sin(x)} = \sin(x)-\frac{\sin^3 x}{3}$$

$u=\sin t$, $du=\cos t\,dt$

$$\Rightarrow \frac{x\left(\sin(x)-\frac{\sin^3(x)}{3}\right)}{1-\cos(x)} = \frac{x}{1-\cos(x)}\cdot\left(\sin(x)-\frac{\sin^3(x)}{3}\right)$$

DEF: Uniforme continuità:

Sia $f: A \to \mathbb{R}$, si dice unif. continua in $E \subseteq A$ se:

$$\forall \varepsilon > 0,\ \exists \delta > 0 \text{ t.c. } |f(x) - f(y)| < \delta \quad \forall x, y \in E,\ |x - y| < \delta$$

TEO. DI Heine-Cantor:

Sia $f:[a,b] \to \mathbb{R}$, $f \in \mathcal{C}^0([a,b])$, allora è unif. continua

DIM:

Supponiamo per assurdo che f non sia unif continua, quindi

$$\exists \varepsilon_0 > 0 \mid \forall \delta > 0 \mid \exists x, y \in [a,b] \text{ con } |x - y| < \delta,\ |f(x) - f(y)| \geq \varepsilon_0$$

Definisco $\delta_n := \frac{1}{n}$

Per ipotesi,

$\exists x, y \in [a,b]$ TALI per cui: $|x_n - y_n| < \frac{1}{n}$, $|f(x_n) - f(y_n)| \geq \varepsilon_0$

Poiché $(x_n)_n$ è contenuta in $[a,b]$, quindi è limitata, allora per B-W,

$$\exists x_{n_k} \to x \in [a,b]$$

Poiché $|x_{n_k} - y_{n_k}| < \frac{1}{n_k} \to 0 \Rightarrow y_{n_k} \to x$

Poiché f è continua in x

$$f(x_{n_k}) \to f(x) \text{ e } f(y_{n_k}) \to f(x)$$

$$\Rightarrow |f(x_{n_k}) - f(y_{n_k})| \to 0$$

Ma per ipotesi dell'assurdo si ha che

$$|f(x_{n_k}) - f(y_{n_k})| \geq 0 \quad \forall k \in \mathbb{N}^+$$

$\Rightarrow$ Contraddizione, quindi f è unif continua

Derivabil. e differen.

DEF: Differenziabilità e retta Tangente

Sia data $f: I \to \mathbb{R}$, definita in $I \subset \mathbb{R}$, $x_0 \in I$

Allora f è differenziabile nel punto x_0 se $\exists m \in \mathbb{R}$ |

$$\lim_{x \to x_0} \frac{f(x) - [f(x_0) + m(x - x_0)]}{x - x_0} = 0$$

e $y = f(x_0) + m(x - x_0)$ è detta retta Tangente al grafico in $(x_0, f(x_0))$

DEF: Derivabilità

Sia $f: I \subseteq \mathbb{R} \to \mathbb{R}$, e $x_0 \in I$, allora f è derivabile se $\exists l \in \mathbb{R}$.

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = l$$

$l = f'(x_0)$ = derivata di f in x_0

TEO: Continuità delle f derivabili

Se f è derivabile in x_0, allora è continua in x_0

Dim: f derivabile $\Rightarrow$

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) \Rightarrow f(x) - f(x_0) = \frac{f(x) - f(x_0)}{(x - x_0)} \cdot (x - x_0)$$

$$\lim_{x \to x_0} f(x) - f(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{(x - x_0)} \cdot (x - x_0)$$

$$\lim_{x \to x_0} f(x) - f(x_0) = \lim_{x \to x_0} l \cdot (x - x_0) = 0$$

$\Rightarrow f(x) = f(x_0) \Rightarrow$ continua